

SUMS OF POWERS OF INTEGERS: A COMPLETE FRAMEWORK FOR  
CLOSED FORMULAS

PETRO KOLOSOV

ABSTRACT.

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## 1. DEFINITIONS AND NOTATIONS

Allow us to define the mathematical objects we utilize throughout the manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], because it is simple and beautiful.

**Definition 1.1** (Multifold sums of powers). *For non-negative integers  $r, n, m$*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We utilize the following notation for central factorials, see also [2, 3, 4]

**Definition 1.2** (Central factorials). *For integers  $n, k$*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where  $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right)$  are falling factorials.

## 2. HISTORICAL CONTEXT

## 3. HELPER LEMMAS

**Lemma 3.1** (Central Newton's formula). *For a function  $f$  defined on integers and arbitrary integers  $x, t$*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where  $\delta^k f(t)$  denotes the  $k$ -th central finite difference

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right),$$

and  $(x-t)^{[k]}$  are central factorials defined by (1.2).

*Proof.* See [2, p. 462], and [5, section 2, eq. (19)], and [3, section 6, eq. (21)]. □

**Lemma 3.2** (Newton's formula for powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

**Lemma 3.3** (Central factorials binomial form). *For integers  $n$  and  $k \geq 1$ ,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

*Proof.* We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left( n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k(k-1)!} \left( n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials  $\frac{(x)_n}{n!} = \binom{x}{n}$ . □

**Lemma 3.4** (Central factorials binomial form decomposition). *For integers  $n, k \geq 0$ ,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \binom{n + \frac{k}{2}}{k} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k}.$$

*Proof.* We have  $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$ . Let  $a = n + \frac{k}{2}$ , then

$$\binom{n + \frac{k}{2}}{k} = \frac{n + \frac{k}{2}}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Thus,

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \binom{n + \frac{k}{2}}{k} - \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[ 2 \binom{n + \frac{k}{2}}{k} - \binom{n + \frac{k}{2} - 1}{k-1} \right] = \frac{1}{2} \left[ \binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} + \binom{n + \frac{k}{2} - 1}{k-1} - \binom{n + \frac{k}{2} - 1}{k-1} \right].$$

Thus, the statement follows. □

**Lemma 3.5** (Newton's formula for powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k} \right] \delta^k t^m.$$

**Lemma 3.6** (Generalized hockey-stick identity). *For integers  $a, b$  and  $j$ ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

*Proof.* We have,  $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left( \sum_{k=0}^b \binom{k}{j} \right) - \left( \sum_{k=0}^{a-1} \binom{k}{j} \right)$ . By hockey stick identity  $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$  yields,

$$\sum_{k=a}^b \binom{k}{j} = \left( \sum_{k=0}^b \binom{k}{j} \right) - \left( \sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

**Lemma 3.7** (Centered hockey-stick identity). *For integers  $n \geq 1$ , and arbitrary integers  $t, k, r$*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

*Proof.* By setting  $j = 1 - t + \frac{k}{2} + r$ , and  $b = n - t + \frac{k}{2} + r$  yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k}.$$

By generalized hockey-stick identity (3.6),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows. □

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4. ORDINARY SUMS OF POWERS

**Proposition 4.1** (Ordinary sums of powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \delta^k t^m \sum_{j=1}^n \left[ \binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k} \right].$$

**Proposition 4.2** (Ordinary sums of powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] \delta^k t^m.$$

For example,

**Example 4.3** (Sums of squares in central differences).

$$\begin{aligned} t = 0 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(1+n) + \frac{\delta^2 t^2}{2} \frac{n(1+3n+2n^2)}{6}, \\ t = 1 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-1) + \frac{\delta^2 t^2}{2} \frac{n(1-3n+2n^2)}{6}, \\ t = 2 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-3) + \frac{\delta^2 t^2}{2} \frac{n(13-9n+2n^2)}{6}, \\ t = 3 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-5) + \frac{\delta^2 t^2}{2} \frac{n(37-15n+2n^2)}{6}, \\ t = 4 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-7) + \frac{\delta^2 t^2}{2} \frac{n(73-21n+2n^2)}{6}. \end{aligned}$$


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**Example 4.4** (Sums of cubes in central differences).

$$\begin{aligned}
t = 0 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(1+n) + \frac{\delta^2 t^3}{2} \frac{n(1+3n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-1+n+4n^2+2n^3)}{24}, \\
t = 1 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-1) + \frac{\delta^2 t^3}{2} \frac{n(1-3n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(1+n-4n^2+2n^3)}{24}, \\
t = 2 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-3) + \frac{\delta^2 t^3}{2} \frac{n(13-9n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-21+25n-12n^2+2n^3)}{24}, \\
t = 3 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-5) + \frac{\delta^2 t^3}{2} \frac{n(37-15n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-115+73n-20n^2+2n^3)}{24}, \\
t = 4 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-7) + \frac{\delta^2 t^3}{2} \frac{n(73-21n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-329+145n-28n^2+2n^3)}{24}.
\end{aligned}$$

It is indeed interesting to notice the connection between the decomposition of ordinary sums of powers (4.2) and Riordan's central factorial numbers of the second kind, that is

**Proposition 4.5** (Central factorial numbers sums). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n + \frac{k}{2} + 1}{k+1} - \binom{1 + \frac{k}{2}}{k+1} + \binom{n + \frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1} \right] k! T(m, k),$$

where  $T(n, k)$  denotes the central factorial numbers of the second kind (in Riordan's notation). See [3, p. 213], and [4].

These central factorial numbers of the second kind  $T(n, k)$  are defined by central Newton's formula for powers (3.2) evaluated in zero, such that

$$n^m = \sum_{k=0}^{\infty} T(m, k) n^{[k]}.$$

Which implies that

$$T(n, k) = \frac{\delta^k 0^m}{k!},$$

by Newton's formula (3.2).

5. REMARK ON CENTRAL FACTORIAL NUMBERS OF THE SECOND KIND

6. GENERALIZED CENTRAL FACTORIAL NUMBERS OF THE SECOND KIND

**Definition 6.1** (Generalized central factorial numbers of the second kind). *For integers  $0 \leq k \leq m$ , and an arbitrary integer  $t$ , the generalized central factorial numbers of the second kind are defined by the relation*

$$n^m = \sum_{k=0}^{\infty} T_t(m, k) \cdot (n - t)^{[k]}.$$

It is also straightforward that

$$T_t(m, k) = \frac{\delta^k t^m}{k!},$$

by Newton's formula (3.2). For example,

| $n/k$ | 0 | 1              | 2 | 3               | 4  | 5              | 6  | 7 | 8 |
|-------|---|----------------|---|-----------------|----|----------------|----|---|---|
| 0     | 1 |                |   |                 |    |                |    |   |   |
| 1     | 0 | 1              |   |                 |    |                |    |   |   |
| 2     | 0 | 0              | 1 |                 |    |                |    |   |   |
| 3     | 0 | $\frac{1}{4}$  | 0 | 1               |    |                |    |   |   |
| 4     | 0 | 0              | 1 | 0               | 1  |                |    |   |   |
| 5     | 0 | $\frac{1}{16}$ | 0 | $\frac{5}{2}$   | 0  | 1              |    |   |   |
| 6     | 0 | 0              | 1 | 0               | 5  | 0              | 1  |   |   |
| 7     | 0 | $\frac{1}{64}$ | 0 | $\frac{91}{16}$ | 0  | $\frac{35}{4}$ | 0  | 1 |   |
| 8     | 0 | 0              | 1 | 0               | 21 | 0              | 14 | 0 | 1 |

**Table 1.** Values of generalized central factorial numbers  $T_0(n, k)$ . See also the sequences [A000000](#), and [A000000](#) in [6].

| $n/k$ | 0 | 1                 | 2   | 3                 | 4   | 5               | 6  | 7 | 8 |
|-------|---|-------------------|-----|-------------------|-----|-----------------|----|---|---|
| 0     | 1 |                   |     |                   |     |                 |    |   |   |
| 1     | 1 | 1                 |     |                   |     |                 |    |   |   |
| 2     | 1 | 2                 | 1   |                   |     |                 |    |   |   |
| 3     | 1 | $\frac{13}{4}$    | 3   | 1                 |     |                 |    |   |   |
| 4     | 1 | 5                 | 7   | 4                 | 1   |                 |    |   |   |
| 5     | 1 | $\frac{121}{16}$  | 15  | $\frac{25}{2}$    | 5   | 1               |    |   |   |
| 6     | 1 | $\frac{91}{8}$    | 31  | 35                | 20  | 6               | 1  |   |   |
| 7     | 1 | $\frac{1093}{64}$ | 63  | $\frac{1491}{16}$ | 70  | $\frac{119}{4}$ | 7  | 1 |   |
| 8     | 1 | $\frac{205}{8}$   | 127 | $\frac{483}{2}$   | 231 | 126             | 42 | 8 | 1 |

**Table 2.** Values of generalized central factorial numbers  $T_1(n, k)$ . See also the sequences [A000000](#), and [A000000](#) in [6].

| $n/k$ | 0   | 1                  | 2    | 3                  | 4    | 5               | 6   | 7  | 8 |
|-------|-----|--------------------|------|--------------------|------|-----------------|-----|----|---|
| 0     | 1   |                    |      |                    |      |                 |     |    |   |
| 1     | 2   | 1                  |      |                    |      |                 |     |    |   |
| 2     | 4   | 4                  | 1    |                    |      |                 |     |    |   |
| 3     | 8   | $\frac{49}{4}$     | 6    | 1                  |      |                 |     |    |   |
| 4     | 16  | 34                 | 25   | 8                  | 1    |                 |     |    |   |
| 5     | 32  | $\frac{1441}{16}$  | 90   | $\frac{85}{2}$     | 10   | 1               |     |    |   |
| 6     | 64  | $\frac{931}{4}$    | 301  | 190                | 65   | 12              | 1   |    |   |
| 7     | 128 | $\frac{37969}{64}$ | 966  | $\frac{12411}{16}$ | 350  | $\frac{371}{4}$ | 14  | 1  |   |
| 8     | 256 | $\frac{6001}{4}$   | 3025 | 3003               | 1701 | 588             | 126 | 16 | 1 |

**Table 3.** Values of generalized central factorial numbers  $T_2(n, k)$ . See also the sequences [A000000](#), and [A000000](#) in [6].

| $n/k$ | 0    | 1                   | 2     | 3                  | 4    | 5               | 6   | 7  | 8 |
|-------|------|---------------------|-------|--------------------|------|-----------------|-----|----|---|
| 0     | 1    |                     |       |                    |      |                 |     |    |   |
| 1     | 3    | 1                   |       |                    |      |                 |     |    |   |
| 2     | 9    | 6                   | 1     |                    |      |                 |     |    |   |
| 3     | 27   | $\frac{109}{4}$     | 9     | 1                  |      |                 |     |    |   |
| 4     | 81   | 111                 | 55    | 12                 | 1    |                 |     |    |   |
| 5     | 243  | $\frac{6841}{16}$   | 285   | $\frac{185}{2}$    | 15   | 1               |     |    |   |
| 6     | 729  | $\frac{12753}{8}$   | 1351  | 585                | 140  | 18              | 1   |    |   |
| 7     | 2187 | $\frac{372709}{64}$ | 6069  | $\frac{53011}{16}$ | 1050 | $\frac{791}{4}$ | 21  | 1  |   |
| 8     | 6561 | $\frac{167943}{8}$  | 26335 | $\frac{35049}{2}$  | 6951 | 1722            | 266 | 24 | 1 |

**Table 4.** Values of generalized central factorial numbers  $T_3(n, k)$ . See also the sequences [A000000](#), and [A000000](#) in [6].

| $t$ | $T_t(2m, 2k)$                                                                | OEIS ID                 |
|-----|------------------------------------------------------------------------------|-------------------------|
| 0   | 1, 0, 1, 0, 1, 1, 0, 1, 5, 1, 0, 1, 21, 14, 1, 0, 1, 85, 147, 30, 1, $\dots$ | <a href="#">A269945</a> |
| 1   | 1, 1, 1, 1, 7, 1, 1, 31, 20, 1, 1, 127, 231, 42, 1, 1, 511, $\dots$          | <a href="#">A394692</a> |
| 2   | 1, 4, 1, 16, 25, 1, 64, 301, 65, 1, 256, 3025, 1701, 126, 1, $\dots$         | <a href="#">A395456</a> |
| 3   | 1, 9, 1, 81, 55, 1, 729, 1351, 140, 1, 6561, 26335, 6951, 266, 1, $\dots$    | <a href="#">A395457</a> |

**Table 5.** Sequences of generalized central factorial numbers  $T_t(2m, 2k)$ .

Thus, we have the formula for sums of powers in terms of generalized central factorial numbers of the second kind.

**Proposition 6.2** (Generalized Central factorial sums). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] k! T_t(m, k).$$

## 7. DOUBLE AND TRIPLE SUMS OF POWERS IN TERMS OF CENTRAL DIFFERENCES

**Proposition 7.1** (Double sums of powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^m \left[ & + \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 \right. \\ & \left. + \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 \right] \delta^k t^m \end{aligned}$$

For example,

**Example 7.2** (Double sum of squares in central differences).

$$t = 0 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(1+n)^2(2+n)}{12},$$

$$t = 1 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-1+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n^2(-1+n^2)}{12},$$

$$t = 2 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-4-3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(10+5n-4n^2+n^3)}{12},$$

$$t = 3 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-7-6n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(32+23n-8n^2+n^3)}{12},$$

$$t = 4 : \quad \Sigma^2 n^2 = \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-10-9n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(66+53n-12n^2+n^3)}{12}.$$

**Proposition 7.3** (Triple sums of powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^3 n^m = \frac{1}{2} \sum_{k=0}^m \left[ & + \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \right. \\ & \left. + \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \right] \delta^k t^m \end{aligned}$$

For example,

**Example 7.4** (Triple sum of squares in central differences).

$$t = 0 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(6+11n+6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(18+45n+40n^2+15n^3+2n^4)}{120},$$

$$t = 1 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-2-n+2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(-2-5n+5n^3+2n^4)}{120},$$

$$t = 2 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-10-13n-2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(58+65n-5n^3+2n^4)}{120},$$

$$t = 3 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-18-25n-6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(198+255n+40n^2-15n^3+2n^4)}{120},$$

$$t = 4 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-26-37n-10n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(418+565n+120n^2-25n^3+2n^4)}{120}.$$

## 8. MULTIFOLD SUMS OF POWERS IN TERMS OF CENTRAL DIFFERENCES

**Proposition 8.1** (Multifold sums of powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k t^m. \end{aligned}$$

For example,

**Example 8.2** (Examples of multifold sums).

$$\begin{aligned}
 t = 0 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} \right. \\
 &\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k 0^m, \\
 t = 1 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} \right. \\
 &\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k 1^m, \\
 t = 2 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} \right. \\
 &\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k 2^m.
 \end{aligned}$$

## 9. MULTIFOLD SUMS OF POWERS IN GENERALIZED CENTRAL FACTORIAL NUMBERS

**Proposition 9.1** (Multifold central factorial sums of powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned}
 \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\
 &\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} k! T_t(m, k).
 \end{aligned}$$

**Example 9.2** (Sums of squares in central factorial numbers).

$$t = 0 : \quad \Sigma^1 n^2 = n \cdot 0 + \frac{1}{2}n(1+n) \cdot 0 + \frac{1}{6}n(1+3n+2n^2) \cdot 1,$$

$$t = 1 : \quad \Sigma^1 n^2 = n \cdot 1 + \frac{1}{2}n(n-1) \cdot 2 + \frac{1}{6}n(1-3n+2n^2) \cdot 1,$$

$$t = 2 : \quad \Sigma^1 n^2 = n \cdot 4 + \frac{1}{2}n(n-3) \cdot 4 + \frac{1}{6}n(13-9n+2n^2) \cdot 1,$$

$$t = 3 : \quad \Sigma^1 n^2 = n \cdot 9 + \frac{1}{2}n(n-5) \cdot 6 + \frac{1}{6}n(37-15n+2n^2) \cdot 1,$$

$$t = 4 : \quad \Sigma^1 n^2 = n \cdot 16 + \frac{1}{2}n(n-7) \cdot 8 + \frac{1}{6}n(73-21n+2n^2) \cdot 1,$$

$$t = 5 : \quad \Sigma^1 n^2 = n \cdot 25 + \frac{1}{2}n(n-9) \cdot 10 + \frac{1}{6}n(121-27n+2n^2) \cdot 1.$$

We may notice that

- For  $t = 0$  the coefficients ... correspond to the second row of the triangle [reference].
- For  $t = 1$  the coefficients ... correspond to the second row of the triangle [reference].
- For  $t = 2$  the coefficients ... correspond to the second row of the triangle [reference].
- For  $t = 3$  the coefficients ... correspond to the second row of the triangle [reference].

**Example 9.3** (Sums of cubes in central factorial numbers).

$$\begin{aligned}
t = 0 : \quad \Sigma^1 n^4 &= n \cdot 0 + \frac{1}{2}n(1+n) \cdot 0 + \frac{1}{6}n(1+3n+2n^2) \cdot 1 \\
&\quad + \frac{1}{8}n(-1+n+4n^2+2n^3) \cdot 0 + \frac{1}{10}n(-2-5n+5n^3+2n^4) \cdot 1, \\
t = 1 : \quad \Sigma^1 n^4 &= n \cdot 1 + \frac{1}{2}n(n-1) \cdot 5 + \frac{1}{6}n(1-3n+2n^2) \cdot 7 \\
&\quad + \frac{1}{8}n(1+n-4n^2+2n^3) \cdot 4 + \frac{1}{10}n(-2+5n-5n^3+2n^4) \cdot 1, \\
t = 2 : \quad \Sigma^1 n^4 &= n \cdot 16 + \frac{1}{2}n(n-3) \cdot 34 + \frac{1}{6}n(13-9n+2n^2) \cdot 25 \\
&\quad + \frac{1}{8}n(-21+25n-12n^2+2n^3) \cdot 8 \\
&\quad + \frac{1}{10}n(18-45n+40n^2-15n^3+2n^4) \cdot 1, \\
t = 3 : \quad \Sigma^1 n^4 &= n \cdot 81 + \frac{1}{2}n(n-5) \cdot 111 + \frac{1}{6}n(37-15n+2n^2) \cdot 55 \\
&\quad + \frac{1}{8}n(-115+73n-20n^2+2n^3) \cdot 12 \\
&\quad + \frac{1}{10}n(298-275n+120n^2-25n^3+2n^4) \cdot 1
\end{aligned}$$

We may notice that

- For  $t = 0$  the coefficients ... correspond to the fourth row of the triangle [reference].
- For  $t = 1$  the coefficients ... correspond to the fourth row of the triangle [reference].
- For  $t = 2$  the coefficients ... correspond to the fourth row of the triangle [reference].
- For  $t = 3$  the coefficients ... correspond to the fourth row of the triangle [reference].

## 10. MULTIFOLD SUMS OF POWERS IN BINOMIAL FORM

**Lemma 10.1** (Multifold sum of zero powers). *For integers  $r \geq 0$  and  $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

*Proof.* (1) Let  $r = 0$ , then we have  $\Sigma^0 n^0 = 1$ , by definition (1.1).

(2) Let  $r = 1$ , then we have  $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$ .

(3) Let  $r = 2$ , then we have  $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$ .

(4) Let  $r = 3$ , then we have  $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$ .

(5) Similarly, for arbitrary integer  $r$ , by hockey stick identity  $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$ , yields

$$\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}.$$

□

**Proposition 10.2** (Multifold binomial sums of powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k t^m \end{aligned}$$

For example,

**Example 10.3** (Examples of multifold binomial sums).

$$\begin{aligned} t = 0 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 0^m, \\ t = 1 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 1^m, \\ t = 2 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 2^m. \end{aligned}$$

## 11. RELATED WORKS

## 12. FRAMEWORK FOR SUMS OF POWERS

## 13. CONCLUSIONS

## REFERENCES

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## 14. MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

| Mathematica Function                                        | Validates / Prints              |
|-------------------------------------------------------------|---------------------------------|
| <code>MultifoldSumOfPowersRecurrence[r, n, m]</code>        | Computes $\Sigma^r n^m$ , (1.1) |
| <code>CentralFactorial[n, k]</code>                         | Computes $n^{[k]}$ , (1.2)      |
| <code>CentralFactorialsBinomialFormDecomposition[20]</code> | Validates Lem. (3.4)            |
| <code>ValidateNewtonsFormulaForPowers[10]</code>            | Validates Lem. (3.5)            |
| <code>ValidateCenteredHockeyStickIdentity[5]</code>         | Validates Lem. (3.7)            |
| <code>ValidateOrdinarySumsOfPowersDecomposition[10]</code>  | Validates Prop. (4.2)           |
| <code>ValidateDoubleSumsOfPowersDecomposition[5]</code>     | Validates Prop. (7.1)           |
| <code>ValidateTripleSumsOfPowersDecomposition[5]</code>     | Validates Prop. (7.3)           |
| <code>ValidateMultifoldSumsOfPowers[3]</code>               | Validates Prop. (8.1)           |